



Medical consequences of the Urals radiation accidents: Comparative analysis

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ARTICLE INFO

Keywords:

The Techa River
East Urals Radioactive Trace
Countermeasures
Leukemia
Solid cancer

ABSTRACT

The paper provides comparative analysis of health effects after radioactive contamination of the Techa River and the 1957 radiation accident at the Mayak Production Association (Mayak PA) in terms of efficiency of performed protective measures. Both radiation accidents took place in the Urals Region, Russia, within a short period of time due to poor-quality of storage technology of liquid radioactive waste. Despite different pathways of radioactive contamination of the environment (air or water), similar long-term combined (external γ - and internal) exposure of the population took place. Countermeasures after the 1957 radiation accident included not only scheduled but also emergency evacuation of people. Scheduled activities on the Techa River were delayed and insufficient and resulted in exposure of local residents at doses greatly exceeding permissible levels. It also led to development of both early deterministic effects and increase in leukemia and solid cancer incidence, and mortality in the long-term period.

Introduction

In the 1950s, two radiation accidents took place at the Mayak Production Association (Mayak PA). It led to radioactive contamination of a part of the Urals region and public exposure. In 1949–1956, Mayak PA exercised operational and accidental releases of liquid radioactive waste (LRW) from different stages of the technological cycle of radiochemical production into the Techa River. The East Urals Radioactive Trace (EURT) was formed on 29 September 1957 as a result of the thermochemical explosion of the LRW storage tank at Mayak PA. The causes of both accidents have been analyzed and described in detail (Burnazyan, 1990; Akleyev and Kisselyov, 2001, 2002). After both accidents, radioactive contamination of the Urals Region was comparable in scale. Total activity of releases into the Techa River was about 115 PBq. The main share of emission activity in 1957 deposited at the Mayak PA industrial site. Total β -activity forming the EURT and exposing the population was about 74 PBq (Aleksakhin et al., 2001; Vasilenko et al., 2019).

Post-accidents countermeasures differed significantly due to a number of reasons, most importantly in levels of preparedness and experience of civil services for accident management arrangement (Burnazyan, 1990; Akleyev and Kisselyov, 2001, 2002; Aleksakhin et al.,

2001). In this respect, comparative analysis of medical consequences after the said accidents with regard to efficiency of performed protective measures is of considerable interest.

Main characteristics of the accident: general and specific features of public exposure

Contamination of the Techa River started in January 1949 and was due to accidental and operational low-level radioactive LRW releases that lasted for almost seven years. Massive process waste releases into the Techa River started in September 1950. Accidental releases took place in the summer and autumn of 1951 amid regular process releases. Accidental releases were due to uncontrolled discharges of high-level waste from storage tanks for LRW. Accidental releases peaked on 8–12 October, 1951. Starting from 28 October 1951 all operational LRW releases were switched to the closed Lake Karachay and that resulted in significant decrease of radionuclide inflow into the Techa River (Degteva et al., 2016).

Radionuclide composition of operational and accidental releases differed significantly. $^{106}\text{Ru}+^{106}\text{Rh}$ (35%), $^{95}\text{Zr}+^{95}\text{Nb}$ (21%), ^{89}Sr (19%) and ^{137}Cs (14%) prevailed in operational discharges. $^{144}\text{Ce}+^{144}\text{Pr}$ constituted 67% of accidental releases. The contribution of $^{95}\text{Zr}+^{95}\text{Nb}$

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(17%) and $^{90}\text{Sr} + ^{90}\text{Y}$ (6%) was significantly lower and that of the other fission products did not exceed 5% (Vasilenko et al., 2019). It should be noted that frequency, isotopic composition, and activity of LRW releases into the Techa River differed greatly in time and that created significant dose rate variability of external gamma and internal exposure. The major share (over 60 %) of relatively long-lived radionuclides such as ^{90}Sr and ^{137}Cs , was discharged into the Techa River in October 1951. It was the most significant contribution to the formation of exposure doses of the Techa residents (Degteva et al., 2016).

The 1957 Mayak PA accident was caused by failure of cooling system of a tank with high-level LRW from radiochemical plant. The explosion of dry nitrate and sodium acetate salts occurred because of evaporation of solutions in the tank. Medium- and long-lived radionuclides were released into the atmosphere with a total activity of about 740 PBq. The major share (about 90%) of activity deposited at Mayak PA. A smaller part of the explosion products (about 74 PBq) rose to an altitude of about 1,000 m and formed a radioactive cloud. Radioactive mixture from the cloud was carried by the wind to the north-east from the explosion site. It led to the formation of the EURT that completed 10–11 h after the explosion (Akleyev et al., 2017). The total activity of the 1957 emission consisted mainly of $^{144}\text{Ce} + ^{144}\text{Pr}$ (66.0 % of total activity), $^{95}\text{Zr} + ^{95}\text{Nb}$ constituted 24.9% of activity, $^{90}\text{Sr} + ^{90}\text{Y} - 5.4 \%$, $^{106}\text{Ru} + ^{106}\text{Rh} - 3.7 \%$, and $^{137}\text{Cs} - 0.1\%$ (Burnazyan, 1990; Akleyev and Kisselyov, 2001, 2002; Aleksakhin et al., 2001).

Discharges into the Techa River and activity of the 1957 emission were of the same magnitude; isotopic composition of releases and emission was almost the same. The accidents differed in pathways of environmental contamination: the Techa River components (floodplain, bottom sediments, aquatic vegetation, ichthyofauna, etc.) were contaminated via water; the EURT territory including settlements and agricultural land – via air. The polluted river with its floodplain was the source of exposure of the Techa residents, and the EURT population was exposed due to radioactively-contaminated territories of the settlements, farmland, forests, lakes, and other environmental factors. Still, the nature of public exposure was similar, which was mainly due to identical living conditions and economic activity of the population. From economic perspective, radioactively contaminated areas of the Techa River and the EURT were typically agricultural. The local population cultivated vegetables, potato, grains (mostly wheat) and produced dairy products and meat (Akleyev and Kisselyov, 2001, 2002).

The residents of the Techa riverside and the EURT settlements were affected by long-term combined (γ -external and internal) exposure. In the first two years after both accidents, outdoor and indoor external exposure predominated in formation of exposure doses to the public. External exposure was caused by an increased γ -background due to short-lived radionuclides in the territory of the settlements and adjacent territories used by the population for economic purposes. Vegetable gardens, pastures, and hayfields were mainly located on floodplain soils and flood meadows of the Techa River. The Techa River and the EURT water bodies were also used for watering vegetable gardens, fishing, swimming, and other household purposes.

In the first years after the accidents, internal exposure was due to short-lived radionuclide intake with locally-produced food products (vegetables, potato, fish, waterfowl meat, milk, etc.). In 1950 and 1951, radionuclide intake by the residents of the Techa settlements was mainly determined by river water, largely used by the locals for drinking and cooking (Akleyev and Kisselyov, 2002). After the 1957 accident, contaminated houses, clothes, household items, domestic animals and fowl, food products and fodder, water supply sources became the sources of radionuclide intake into human bodies (Akleyev and Kisselyov, 2001). ^{144}Ce , ^{95}Zr and ^{106}Ru have largely contributed to the formation of doses of internal exposure of gastrointestinal tract. Inhalation of radionuclides contributed to internal doses significantly less (Aleksakhin et al., 2001). After the decay of short-lived γ -emitting radionuclides, ^{90}Sr contributed to long-term internal exposure of the population residing along the Techa River and the EURT, mainly due to the

consumption of cow milk (Akleyev and Kisselyov, 2001, 2002).

Measures of the radiation protection of the population

Countermeasures implemented on the Techa River and EURT are analyzed in a number of publications (Burnazyan, 1990; Akleyev and Kisselyov, 2001, 2002; Aleksakhin et al., 2001). It should be noted that after the Techa River contamination, protective countermeasures were insufficient and implemented with delay. Emergency protective activities were not performed on the Techa River due to lack of scientifically justified criteria for radiation protection. Scheduled activities included resettlement; hydrotechnical activities (construction of the Techa Cascade of the Reservoirs, well and water-supply construction); restrictive measures on the use of the Techa River and its floodplain for non-resettled residents; arrangement of radiation monitoring and medical follow-up of the population (Table 1) (Akleyev and Kisselyov, 2002).

Priority measures to protect the population of the Techa settlements included the limitation of LRW discharges into the river starting the end of 1951 and the introduction of official ban on the use of the river for drinking and household purposes. Since 1953, a restrictive sanitary protected zone (SPZ) has been established to restrict the access to the river for the population, cattle, and water fowl. The river and its floodplain were separated by a barbed wire fence in the vicinity of the settlements. Outside the settlements the territory was planted with trees and delineated by ditch. The Techa Cascade of Reservoirs (TCR) was constructed to isolate the most heavily contaminated upper reaches of the river. It included drain water ponds constructed with the help of dams and by-pass canals (Akleyev and Kisselyov, 2002).

From 1954 to 1960, over 7000 Techa riverside residents were resettled from 19 villages, which constituted about 30% of total number of residents. The population was resettled gradually starting from the settlements located in the upper reaches of the river (Akleyev and Kisselyov, 2002; Aleksakhin et al., 2001).

A decrease in the effect of radiation exposure on the remaining riverside settlement population was supposed to be achieved by introducing sanitary protected regime (SPR). Drinking and household water use restrictions were imposed on the non-resettled population (vegetable gardens watering, clothes washing, water fowl breeding, vegetable and potato growing in the floodplain, etc.). Instead, the population was provided with clean water sources. Wells and water pipes were built to use groundwater.

It is necessary to note that SPR for non-resettled residents came into operation with delay. For example, the ban on drinking river water was introduced only 3-4 years after the onset of LRW discharges. However, since well and water-pipe construction did not finish until 6-7 years after

Table 1
Protective measures in the Techa River basin and in the EURT territory.

Measures	Techa River	EURT
Emergency	Not implemented	Evacuation of 1100 persons
Scheduled	Sanitary-protected zone, 80 km ² (SPZ)	SPZ, 700 km ²
	Resettlement of 7,338 persons*	Resettlement of 10,580 persons
	Sanitary-protective regime (SPR)	SPR
	Hydrotechnical activities	Radioactive decontamination of settlements and agricultural areas Quality inspection of food products and fodder with their partial replacement Restructuring and conversion of farming enterprises
	Radiation monitoring	Radiation monitoring
	Medical follow-up	Medical follow-up

*—Additionally, residents of Muslyumovo village were resettled in 2006–2012 due to high social and psychological tension.

the introduction of the ban on river water use, the population had to partially use river water for cooking and other household needs (Akleyev and Kisselyov, 2002).

Conforming to restrictive requirements of the SPR was the most difficult of all protective measures for the population. These measures were frequently violated (fishing and swimming in the river, as well as hay making in the alienated floodplain, but the most frequent one was cattle pasturing in the alienated riverside floodplain). In all villages, a large number of water fowl (geese and ducks) was kept on the river in summers. For a long period of time, one of the most important reasons of non-observance of SPR by the residents was absence of official information about radioactive contamination of the river. The population did not know about the restrictions reasons and the consequences of using the river and its floodplain. Other important factors of non-compliance were lack of "clean" areas for agricultural activities and difficulty in changing the lifestyle of rural residents who used the river for various economic and household needs for many years (Akleyev and Kisselyov, 2002; Aleksakhin et al., 2001).

The first calculations of the EURT levels and boundaries of contamination were made on the first day after the accident. Activity and isotopic composition of the 1957 emission were assessed on the first days after the accident by analyzing the passport data of solutions entering the storage tank. Decisions on countermeasures were based on environmental contamination data and predictive estimates of the population exposure doses.

Protective measures (Table 1) included both emergency and scheduled measures. Emergency activities program to mitigate the accident consequences was approved as early as 2 October, 1957. It included evacuation of the population from the settlements nearest to the explosion site and determination of deposition and boundaries of the contaminated area for the SPZ establishment. Residents of Berdyanish, Satlykovo, and Galikaev villages closest to the explosion site were evacuated within 10 days after the accident. All village residents were decontaminated with a change of clothes. The evacuated residents were provided with new dwellings. Houses and outbuildings were completely destroyed in these villages, and farm livestock were slaughtered and buried (Burnazyan, 1990; Akleyev and Kisselyov, 2001).

In other EURT settlements, radiation monitoring and food quality inspection to partially replace contaminated products began in autumn of 1957 to reduce internal exposure. In the first months after the accident, quality inspection was difficult due to the fact that officially the values of permissible contamination levels were not stated yet. Temporary maximum permissible levels of daily intake of radionuclides were adopted on May 7, 1958. Absence of a territorial radiological service in Chelyabinsk and Sverdlovsk regions hampered radiation monitoring significantly. Mayak PA radiological laboratory alone evaluated the content of radionuclides in food and fodder during this period. The results of the measurements showed that for residents of some settlements radionuclide intake exceeded permissible levels (Akleyev and Kisselyov, 2001; Akleyev et al., 2017).

In October–December 1957, repeated assessment of the EURT contamination levels and boundaries, made it possible to clarify the boundaries of the pollution zone. At the beginning of 1958, the SPZ was established and residence and any economic activity were prohibited. ^{90}Sr deposition of 74 kBq/m^2 was recognized as maximum for population safety and was adopted as the official border of the EURT. Additional evacuation of the population from the SPZ territory was routinely performed. There were 22 settlements with more than 10 thousand people in SPZ. Seven villages (2280 people) were resettled 250 days after the accident, and twelve villages (8300 people) – 330,670 days after the accident (Burnazyan, 1990; Akleyev and Kisselyov, 2001; Aleksakhin et al., 2001).

The efforts were aimed at reducing external exposure doses and the annual intake of ^{90}Sr and ^{137}Cs in the settlements located in the contaminated EURT territories with ^{90}Sr deposition of less than 74 kBq/m^2 . All protective measures were implemented in an attempt to

minimize adverse public health effects, but economic rationale of countermeasures was not assessed (Burnazyan, 1990; Akleyev et al., 2017). The countermeasures included decontamination of settlements and agricultural land mainly by plowing. Restrictions prohibited to use the territories with ^{90}Sr deposition of more than 74 kBq/m^2 for pastures and hayfields.

Developing and managing food contamination limits enabled to reorganize and repurpose local food businesses. Small farms were reorganized into larger ones, specializing in the production of beef and pork, with special conditions for keeping and feeding animals. The use of natural forage lands was limited. Thanks to efforts of specialized farms in the 1960s, the content of ^{90}Sr in beef, the main product, decreased by a factor of 2–7. Grain and vegetable production was prohibited, and milk production was limited. Expansion of farms significantly facilitated radiation monitoring of agricultural products (Akleyev and Kisselyov, 2001; Aleksakhin et al., 2001). These measures allowed creating safe living conditions for the EURT population. However, due to non-uniform radioactive contamination of the EURT and lack of "clean" pastures and hayfields, the population continued to partially use the territory of the SPZ (Akleyev and Kisselyov, 2001).

Main characteristics of TRC and EURTC

LRW discharges caused contamination of the Techa - Iset - Tobol - Ob river system, but the worst radiation situation was observed in the Techa River basin where about 24,000 people resided at the time of the onset of LRW discharges and about 6000 more people arrived or were born in the riverside settlements between 1952 and 1960. 270,000 people resided in 217 villages of about $20,000 \text{ km}^2$ of the EURT territory, where ^{90}Sr deposition was $\geq 3.7 \text{ kBq/m}^2$.

Mobile teams of physicians started medical examinations of the Techa residents 18 months after the discharges, whereas examinations of the EURT population began in the first days after the accident. Until now, specialized medical follow-up of the Techa riverside and EURT residents has been performed by the Clinical department of the Urals Research Center for Radiation Medicine (URCRM) of the Federal Medical Biological Agency. At first, people undergo screening at the place of their residence performed by mobile teams of the URCRM physicians; then they are examined at the URCRM outpatient clinic; and lastly in a hospital setting to verify the diagnosis (Akleyev, 2021a).

Medical-dosimetric database of the URCRM stores data of exposed population medical examinations and accidents analysis (Akleyev and Kisselyov, 2002). The Database comprises results of medical and dosimetric follow-up of the cohort members, information on vital status, residence history, non-radiation carcinogenic risk factors (sex, ethnicity, fact of evacuation, family history, for limited number of population - education, employment, smoking data, body-mass index, number of relatives with cancer) and other factors. The information is regularly updated using common information sources for both cohorts (Akleyev and Kisselyov, 2001, 2002).

Creation of the exposed population registry was started in 1967. The Techa River Cohort (TRC) and the East Urals Radioactive Trace Cohort (EURTC) were established to analyze health effects of chronic exposure in the residents of the Techa and the EURT settlements. The TRC consisted of 29,730 people born before 1950 and those who permanently resided in the Techa riverside villages in 1950–1960 (Akleyev and Kisselyov, 2002). The EURTC included 21,427 residents from 22 resettled and 14 non-resettled settlements, located close to the resettlement border in the proximal part of the EURT, who were born before the accident and resided in the EURT territory before resettlement or until 31 December, 1959 (Akleyev et al., 2017). The main characteristics of the cohorts are presented in Table 2.

Analysis of the main characteristics of TRC and EURTC indicates their sufficient comparability in terms of size, follow-up period, sex ratio, ethnicity, and age structure of the population. Women predominated in both cohorts at the time of exposure, and there was a large

Table 2

Main characteristics of the studied cohorts.

Parameter	TRC	EURTC
Size of the cohort	29,730	21,427
Follow up period, Years*	1950–2007	1957–2006
	58	50
Number of person-years under follow up	927,743	458,131
Percentage of females, %	58	56
Ethnic composition, %:		
Tartars and Bashkirs	20.0	39.7
Slavs	80.0	60.3
Percentage of people under 20 years old at the onset of exposure, %	43	42

Note:

* - the data are given as of the last publication (Krestinina et al., 2017).

proportion of children and young people. It should be noted that the EURTC includes about 1,500 people previously resettled from the Techa riverside settlements where they were exposed to higher doses relative to those received following the 1957 accident.

The follow-up of the vital status and health status of the cohort members has been carried out in the catchment area that slightly differed for cancer incidence and mortality. Cancer incidence catchment area included Krasnoarmeysky, Kunashaksky, Argayashsky, Kaslinsky and Sosnovsky areas of the Chelyabinsk regions. For the population of these areas the information on cancer incidence was available for the whole follow-up period. This catchment area also included the cities of Chelyabinsk and Ozyorsk where the exposed people mainly migrated to. The mortality catchment area included the entire Chelyabinsk and Kurgan regions. Table 3 presents vital status in the compared cohorts (Krestinina et al., 2017).

A proportion of deceased cohort members predominates in both cohorts and constitutes 58% in the TRC and 37% - in the EURTC. A bigger proportion of deceased people in the TRC was due to an older age of the members of this cohort. By the end of the follow-up period, the youngest TRC members reached the age of 58 years, whereas in the EURTC - 50 years. The proportion of alive persons in the TRC catchment area is 19%, and 27% – in the EURTC. The proportion of those who were excluded from the analysis before the end of the follow-up period was 23% and 35%, respectively. A significant part of those lost to follow up in both cohorts (about 16%) are people who migrated outside the catchment area. Migration was mainly associated with the departure of young people from rural areas to get married, study or work, serve in the army, and did not depend on the dose received. A relatively high proportion of dropouts reduces the statistical power of the study but should not lead to a bias in the risk value, since we use censored data, i.e. for each person who left the catchment area, the exact date of the departure is known. After that date both cancer cases and person-years are not included in the analysis.

Table 3

Vital status in the cohorts under comparison

Vital status	TRC		EURTC	
	Number of persons	%	Number of persons	%
Alive in the Catchment area	5684	19	5731	27
Deceased	17,307	58	8016	37
With known cause of death	15,763	91	7164	89
Lost to follow up	6739	23	7680	35
Out of them: Migrants	4696	16	3511	16
Status unknown	2043	7	4169	19
Total	29,730	100	21,427	100

Dosimetric estimates of protective measures efficiency

The population of the Techa riverside and EURT settlements has been chronically exposed to both external γ - and internal radiation due to the intake of uranium fission products. Long-term and significant ^{90}Sr ingress with local food, mainly with milk, caused predominant exposure of red bone marrow (RBM) in the members of both cohorts (Akleyev and Kisselyov, 2001, 2002). As it is seen in Fig. 1, the dynamics of annual RBM doses in the cohort members had a typical accidental character. The highest dose rates were noted during the period of maximum discharges into the Techa River in 1950–1956, and at the EURT - in 1957. In subsequent years, gradual decrease in the annual RBM doses was noted.

The intake of ^{90}Sr determined internal exposure of the Techa and EURT residents for a long time. Its content in the body can be considered as an indicator of the effectiveness of protective countermeasures with regard to internal exposure. Monitoring of ^{90}Sr body burden in the residents of radioactively contaminated areas was performed during lifetime using whole body counter (SICH-9.1) and autopsy data. Radiochemical method was used to measure ^{90}Sr -body burden in bone samples in 246 deceased TRC members and 1,646 EURT residents, while 10,174 and 369 people, respectively, were measured with the use of spectrometry method (Degteva et al., 2018). The results of ^{90}Sr measurements in the cohort members are shown in Fig. 2.

Although the introduction of a SPR led to a decrease in ^{90}Sr intake with milk by the Techa residents by 6.7 times in the middle and 3 times in the distant part of the river (Akleyev and Kisselyov, 2002), the content of ^{90}Sr was high and about two orders of magnitude higher than similar values for the EURT residents.

Individualized organ dose assessments were performed using the Techa River Dosimetry System - TRDS-2016D (Degteva et al., 2019). Table 4 shows the doses to RBM and stomach that were used to assess the radiation risk of leukemias and solid cancers in the compared cohorts. Their mean, median, and maximum values indicate significant excess of the absorbed dose to RBM in the TRC members as compared to the EURTC. Stomach doses were comparable in both cohorts.

The range of individual values of the absorbed RBM dose in the Techa residents was rather wide, including high doses. The maximum absorbed doses to RBM were observed in permanent residents of the upper Techa riverside settlements, which were located close to the site of LRW releases. In the EURTC, the RBM doses were significantly lower: they did not exceed 100 mGy in 90% of the cohort members.

The analysis of the dose averted at gradual resettlement of the EURT residents showed that this drastic protective measure was effective due to the timing of resettlement. Maximum values of the averted doses of the population were observed during resettlement in the first 10 days after the accident. Evacuation decreased significantly the doses of external γ -exposure and halted dietary intake of relatively short-lived radionuclides (mainly ^{144}Ce). Later resettlement of the EURT residents (250–670 days after the accident) did not result in a significant decrease in the doses of external and internal exposure to the gastrointestinal tract since their main share was formed in the first months after the accident. Thus, resettlement 670 days after the explosion averted only 10% of collective effective dose and 20% of absorbed dose to RBM (Aleksakhin et al., 2001).

Medical consequences of the accidents

Deterministic effects of radiation exposure, namely chronic radiation syndrome (CRS) and pronounced reactions of hematopoietic, immune, and central nervous systems were registered only in the Techa residents. They were mainly observed in 1952–1958 among the residents of the settlements located closest to the site of LRW discharges (Akleyev and Kisselyov, 2002). The exact number of CRS cases among the Techa riverside residents remains unknown, since not all exposed people underwent medical examinations. The residents were diagnosed with CRS of mild severity, which can be diagnosed only on the basis of long-term

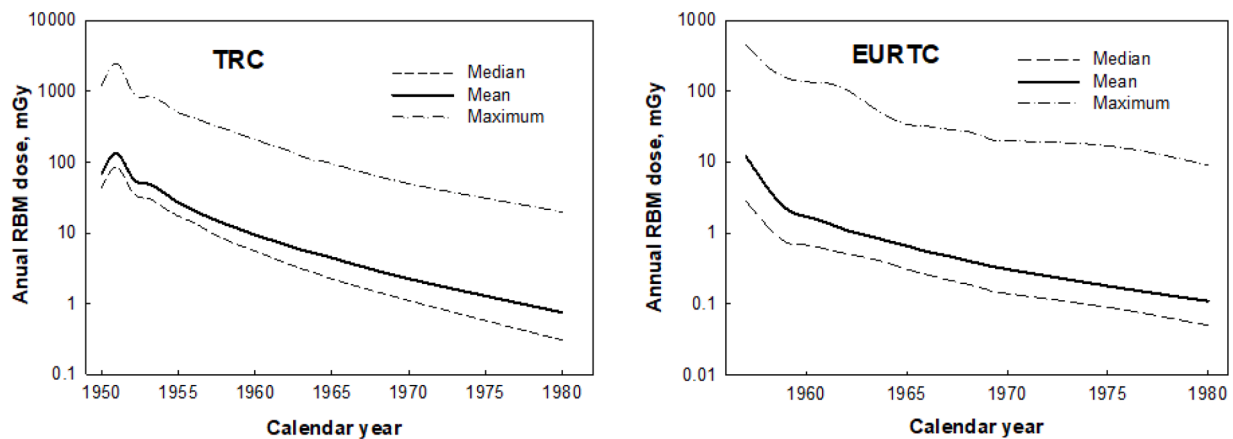


Fig. 1. Annual RBM doses for the members of the TRC and EURTC.

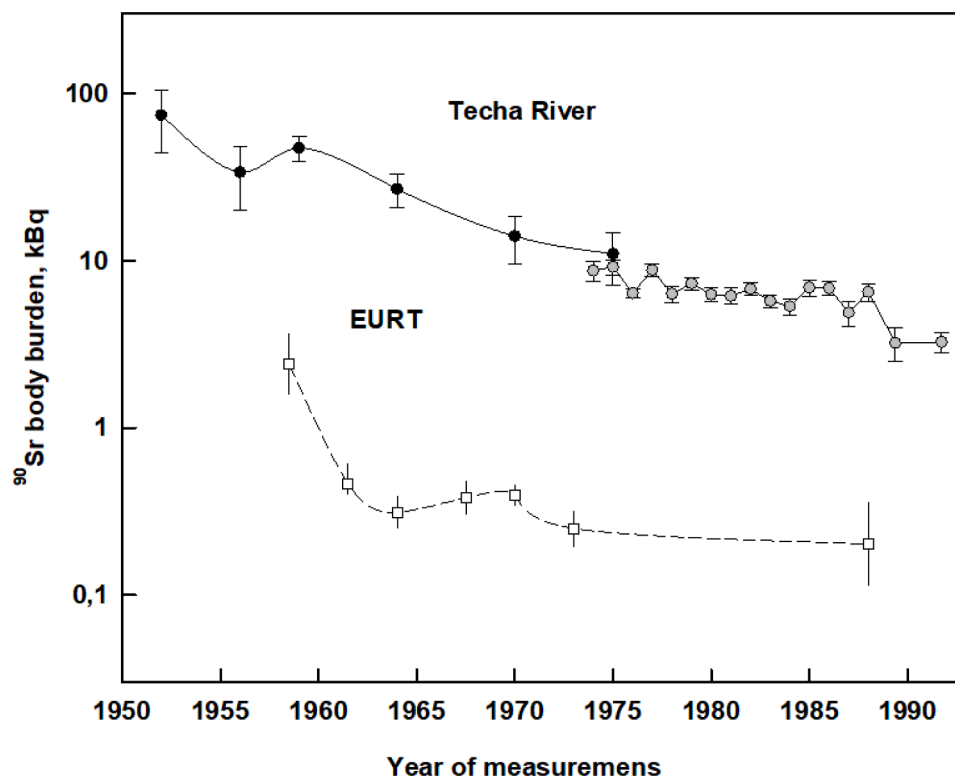


Fig. 2. ^{90}Sr -body burdens measured for adult residents of the Techa River and EURT. The upper curve corresponds to the inhabitants of the Techa riverside communities less than 100 km from the release site (black color shows postmortem measurements, and gray color – *in vivo* measurements). The lower curve corresponds to the EURT residents who lived less than 100 km from the accident site. Vertical bars indicate standard errors of mean.

regular medical follow-up (Akleyev, 2014, 2021b).

Hematopoietic disorders such as cytopenia, were often detected in the residents of the Techa riverside settlements even during the first medical examination. The most pronounced changes in hematopoiesis were observed in 1951–1953 when maximum dose rates of external γ -exposure were registered. They manifested as granulocytopenia, lymphocytopenia and thrombocytopenia. Such neurological disorders as vegetative dysfunction, fatigue, and organic damage to nervous system appeared later and persisted longer than changes in hematopoiesis (Akleyev and Kisselyov, 2001, 2002). The fact that CRS and reactions of critical body systems predominated in the residents of the upper Techa riverside settlements pointed at the prevailing contribution of external γ -exposure to their development, against which no emergency countermeasures were performed in the villages of the Techa River basin.

Due to timely evacuation of residents of the settlements closest to the exposure site, no cases of acute or chronic radiation syndrome were registered in the EURT population. Mild hematopoietic reactions were observed in some EURT residents only during the first year after the accident. They had a transient moderate decrease in the number of leukocytes, neutrophils and thrombocytes in the peripheral blood (Akleyev et al., 2017).

The rates of leukemia and solid cancer incidence and mortality were considered as long-term medical criteria of the effectiveness of protective measures. Excess relative risk (ERR) of solid cancers in the TRC and EURTC is presented in Table 5 (Schonfeld et al., 2013; Davis et al., 2015; Akleyev et al., 2017; Krestinina et al., 2017; Degteva et al., 2018; Silkin et al., 2020). ERR of solid cancers (excluding skin cancer and osteosarcoma) was increased according to both incidence and mortality data in

Table 4

Main characteristics of the exposure doses to RBM and stomach in residents of the Techa River and EURT settlements

Absorbed dose	TRC	EURTC
RBM, Gy		
Mean value	0.36	0.12
Median	0.20	0.01
Maximum value	7.92	2.04
Stomach, Gy*		
Mean value	0.06	0.03
Median	0.02	0.002
Maximum value	1.13	0.69

Note:

* the provided doses to the stomach are used in the assessment of radiation risk of incidence and mortality from all solid cancers excluding osteosarcoma and skin cancer.

Table 5

Excess relative risk (linear model) of malignant tumors in the Urals cohorts of exposed population

Cohort	ERR/100 mGy(95% CI)	p	ATR*, % (excess cases)
TRC, incidence			
1956-2007; 1,933 cases	0.09 (0.02; 0.16)	0.008	3.6 (69)
EURTC, incidence			
1957-2014; 1,609 cases	0.05 (0.01; 0.10)	0.02	2.3 (37)
1957-2014; 1,366 cases**	0.16 (-0.06; 0.42)	0.15	1.7 (23)
TRC, mortality			
1950-2007; 2,303 cases	0.06 (0.004; 0.13)	0.03	2.2 (50)
EURTC, mortality			
1957-2006; 1,039 cases	0.06 (0.001; 0.13)	0.046	2.4 (25)
1957-2006; 886 cases, **	0.15 (-0.21; 0.63)	0.44	1.6 (14)

Note:

* – attributable risk, excess number of cases are given in parentheses;

** – ERR in the EURT subcohort with the exclusion of people previously exposed on the Techa River.

the TRC and EURTC members.

ERR values of incidence and attributable risk in the TRC were slightly higher than those in the EURTC whereas ERR values of solid cancer mortality were comparable in both cohorts. However, ERR of both solid cancer incidence and mortality in the EURTC becomes statistically insignificant when the EURTC members who used to live in the Techa riverside settlements during the LRW releases (and thus were affected by radiation exposure in two radiation situations) were excluded from the analysis.

In both cohorts, a dose dependence of solid cancer incidence and mortality is well described by a linear model (Akleyev et al., 2017; Krestinina et al., 2017; Silkin et al., 2020). Attributable cancer risk is 0.7-3.6%.

An increased RBM dose-dependent incidence risk for all forms of leukemia as well as leukemias excluding chronic lymphocytic leukemia (CLL) was noted only in the TRC (Table 6). No statistically significant increase in ERR of leukemias with an increase of RBM dose was observed in the EURTC (Krestinina et al., 2019).

High values of attributable leukemia risk (especially with CLL excluded) in residents of the Techa riverside settlements in comparison with solid cancer were largely due to internal ⁹⁰Sr exposure of RBM.

Assessment of the collective effective dose in the compared cohorts was performed on the basis of the recommendations of the International Commission on Radiological Protection (ICRP, 2007) with the use of individual organ doses. Collective effective dose was 2980 person Sv in

Table 6

Excess relative risk (linear model) of leukemias in the Urals cohorts of exposed population

Leukemias	ERR/100 mGy (95% CI)	p	ATR, % (excess cases)
TRC, 1953-2007			
All leukemias; 99 cases	0.12 (0.04; 0.24)	<0.001	30 (29)
Leukemias excluding CLL; 72 cases	0.22 (0.08; 0.54)	<0.001	47 (34)
EURTC, 1957-2009			
All leukemias; 37 cases	0.10 (nf <0; 0.37)	0.14	10 (4)
Leukemias excluding CLL; 29 cases	0.12 (nf <0; 0.45)	0.17	11 (3)

Note: nf (not found) – the limit of the confidence interval could not be calculated by the program.

the TRC, and 340 person Sv – in the EURTC. Calculations prove a much greater long-term damage to public health from LRW discharges into the Techa River than after the 1957 accident.

Conclusion

Efficiency of performed protective measures was evaluated by assessing nature of exposure and specifics of exposed population (size, age, sex) as well as analyzing health effects of LRW discharges into the Techa River and the 1957 accident at the Mayak PA. Emergency protective measures were not implemented on the Techa River, whereas scheduled countermeasures were insufficient and delayed, which caused chronic radiation exposure of the population at a wide dose range with the predominant exposure of RBM. Partial resettlement of the population began only 5 years after the onset of releases and continued until 1961. For several years, the population was not provided with clean water for cooking and household purposes because of the delay in well and water pipe construction. There were not enough clean areas for vegetable gardens, pastures and hayfields. Since no official information about the accident was available, the local population often breached the river water use ban for household purposes (watering vegetable gardens, washing clothes, water fowl breeding, vegetable and potato growing in the floodplain of the river, etc.). All this resulted in radionuclide intake through river water and local food products consumption and in external exposure at doses significantly exceeding permissible levels. The scheduled protective measures implemented after the stop of discharges (SPZ establishment, TCR construction and others) did not enable to significantly reduce the public exposure dose due to their delayed implementation.

Taking into account the Techa River radioactive contamination experience, protective measures on the EURT began right after the accident and were more extensive. Evacuation of the population from the EURT settlements adjacent to the accident site turned out to be highly effective. In the first ten days after the accident, the residents of the adjacent settlements were evacuated. According to the calculations (Burnazyan, 1990), they could have received doses from external γ -exposure up to 1 Gy and higher. This made possible to prevent acute and chronic radiation syndromes.

Timing of countermeasures largely determines their effectiveness. Delay in the implementation of protective measures significantly reduces their effectiveness. Thus, the averted dose decreased significantly in the EURT population resettled 250-670 days after the accident. Resettlement 670 days after the accident averted only 10% of effective dose and 20% of absorbed dose to the RBM. Thus, delayed post-accident resettlement of the population from radioactively-contaminated territories was not reasonable enough and could have a negative effect on their quality of life

Soon after the accident, SPZ was established, radiation monitoring and pollution control were arranged. Scheduled protective measures

included additional population resettlement, decontamination of the settlements and agricultural areas, restructuring and converting food producing businesses, and others. These measures enabled to lower ^{90}Sr intake and significantly reduce internal doses. Thus, the main long-term factor of internal exposure - ^{90}Sr content, the marker radionuclide - was two orders of magnitude lower in the EURT population than in those of the Techa River. As a result, the countermeasures made it possible to significantly reduce doses of both internal and external exposure of the EURT population, as compared to those in the TRC members, and to minimize the adverse health effects of the EURT population.

Thus, comparative analysis of the results of long-term monitoring of ^{90}Sr content, exposure doses and health status of accidentally exposed residents of the Techa riverside and EURT settlements indicate a significantly greater efficiency of the protective measures taken after the 1957 accident. The non-use of emergency countermeasures in the settlements in the proximal part of the Techa River resulted in the exposure of the population in 1950 – 1960 at doses sufficient for the development of CRS and tissue reactions of hematopoiesis, systemic immunity and nervous system. Increased radiation risk of leukemia and solid cancer in the population is also a secondary indicator of low effectiveness of countermeasures taken after radioactive contamination of the Techa River. This statement is also confirmed by the fact that the collective effective exposure dose, which characterizes the damage to health mainly from solid cancers and leukemias, was about an order of magnitude higher in the TRC members than that in the EURT.

Financial support

This work has been carried out under the Contract N^o 27.501.19.2 dated 08 July 2019 in the framework of Russian Federal Targeted Program “Provision of nuclear and radiation safety for the period 2016–2020 and for the period up to 2030”.

CRedit authorship contribution statement

Alexander V. Akleyev: Project administration, Conceptualization, Writing – original draft, Writing – review & editing. **Marina O. Degteva:** Writing – original draft, Writing – review & editing. **Lyudmila Yu. Krestinina:** Writing – original draft, Writing – review & editing.

Declaration of Competing Interest

The authors declare no conflicts of interest regarding the publication of this manuscript.

Acknowledgments

The authors are grateful to the staff of the Medical-dosimetric database department (head of the department – N.V. Startsev) for assistance in conducting the follow-up of cohort members vital status, to

the personnel of the Clinical Department (head doctor – V.N. Kovalenko) for the long-term follow-up of the exposed population health status, and the employees of the biophysics (head of the laboratory – M.O. Degteva) and epidemiological laboratories (head of the laboratory – L.Yu. Krestinina) for participating in data collection and preparation.

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